Project Initialization and Planning Phase

Date	15 March 2024
Team ID	SWTID-2026-5339
Project Title	EduGenie – Google Gemini Powered Learning Assistant
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) Report

The proposed solution aims to transform the learning experience by leveraging Google Gemini 1.5 Pro and FastAPI to create an AI-powered learning assistant. The system provides instant answers to academic questions, explains complex concepts in simple language, summarizes lengthy study materials, generates quizzes, and recommends personalized learning paths. This intelligent platform improves learning efficiency, enhances student engagement, and supports self-paced education.

Project Overview	
Objective	The primary objective of this project is to develop an AI-powered learning assistant that helps students learn more effectively by providing intelligent question answering, concept explanations, automatic summarization, quiz generation, and personalized learning recommendations using Google Gemini AI.
Scope	The project focuses on developing a web-based intelligent learning platform that integrates Google Gemini AI with FastAPI to provide personalized educational assistance. The application enables students to interact with AI through a user-friendly interface and supports multiple learning activities within a single platform.

Problem Statement	
Description	Students often struggle to understand difficult concepts, search across multiple platforms for study materials, create quizzes manually, and organize learning resources effectively. Traditional learning methods lack personalized assistance and require significant time to gather relevant educational content.
Impact	Addressing these challenges will improve learning efficiency, reduce study time, increase student engagement, provide personalized educational support, and enhance overall learning outcomes through AI-powered assistance.

Proposed Solution	
Approach	Develop a modular AI-powered web application using FastAPI, Google Gemini 1.5 Pro, HTML, CSS, and Jinja2. The system processes user requests through REST

	APIs and generates intelligent educational responses using prompt engineering and Google's Gemini model.
Key Features	• AI-powered Question Answering System • Topic Explanation Module • Text Summarization Module • Automatic Quiz Generation (MCQs) • Personalized Learning Path Recommendation • FastAPI REST API Integration • Responsive Web Interface • JSON-based Communication • Error Handling and Input Validation

Resource Requirements

Resource Type	**Description**	**Specification / Allocation**
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	Intel Core i5/Ryzen 5 Processor (or above)
Memory	RAM specifications	8 GB RAM
Storage	Disk space for data, models, and logs	256 GB SSD or above
Software		
Framework	Python frameworks	FastAPI, Uvicorn
Libraries	Additional libraries	google-generativeai, FastAPI, Jinja2, HTML, CSS
Development Environment	IDE	Visual Studio Code / PyCharm
AI Model	Large Language Model	Google Gemini 1.5 Pro
Data		
Input Source	Type of input data	User Questions, Topics, Study Materials (Text Input)
Output Format	Response format	JSON Response through REST APIs